

## 1 Introduction

Changing bases is a fundamental technique in linear algebra that allows us to simplify problems by representing vectors in more convenient coordinate systems. A change of basis corresponds to switching from one ordered basis of a vector space to another, and it is implemented using matrix multiplication.

In  $\mathbb{R}^2$ , for example, instead of using the standard basis  $\{\mathbf{e}_1, \mathbf{e}_2\}$ , we may use a different basis that better suits the geometry or physics of a problem, such as a tangent–normal frame.

## 2 Coordinates in $\mathbb{R}^2$

### Definition 2.1: Standard Coordinates

Any vector  $\mathbf{x} \in \mathbb{R}^2$  can be written as

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2,$$

where  $(x_1, x_2)^T$  are called the coordinates of  $\mathbf{x}$  with respect to the standard basis  $\{\mathbf{e}_1, \mathbf{e}_2\}$ .

### Definition 2.2: Coordinates with respect to a basis

Let  $\{\mathbf{y}, \mathbf{z}\}$  be a basis for  $\mathbb{R}^2$ . Then every vector  $\mathbf{x} \in \mathbb{R}^2$  can be written uniquely as

$$\mathbf{x} = \alpha\mathbf{y} + \beta\mathbf{z}.$$

The vector  $(\alpha, \beta)^T$  is called the coordinate vector of  $\mathbf{x}$  with respect to the ordered basis  $[\mathbf{y}, \mathbf{z}]$ .

### Remark 2.1: Ordering of a basis

The order of basis vectors matters. If the ordered basis is changed from  $[\mathbf{y}, \mathbf{z}]$  to  $[\mathbf{z}, \mathbf{y}]$ , then the coordinate vector changes from  $(\alpha, \beta)^T$  to  $(\beta, \alpha)^T$ .

## 3 Transition Matrices in $\mathbb{R}^2$

### 3.1 Constructing a transition matrix

Suppose we are given a new basis

$$\mathbf{u}_1 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

We form the matrix

$$U = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}.$$

### Definition 3.1: Transition Matrix

Let  $U = (\mathbf{u}_1 \ \mathbf{u}_2)$  be the matrix whose columns are basis vectors expressed in the standard basis. Then:

$$\mathbf{x} = U\mathbf{c}$$

maps coordinates  $\mathbf{c}$  in the  $\{\mathbf{u}_1, \mathbf{u}_2\}$ -basis to standard coordinates.

## 3.2 Changing from new basis to standard basis

If

$$\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}, \quad \text{then} \quad \mathbf{x} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2.$$

Expanding:

$$\mathbf{x} = \begin{pmatrix} 3c_1 + c_2 \\ 2c_1 + c_2 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}.$$

## 3.3 Inverse transformation

To recover coordinates in the  $\{\mathbf{u}_1, \mathbf{u}_2\}$  basis from standard coordinates:

$$\mathbf{c} = U^{-1}\mathbf{x}.$$

### Definition 3.2: Inverse Transition Matrix

If  $U$  is a transition matrix from a basis  $B$  to the standard basis, then  $U^{-1}$  is the transition matrix from the standard basis back to  $B$ .

# 4 General Change of Basis in Vector Spaces

## 4.1 Coordinate vectors

### Definition 4.1: Coordinate vector in a general vector space

Let  $V$  be a vector space with ordered basis

$$E = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}.$$

Then every vector  $\mathbf{v} \in V$  can be written uniquely as

$$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_n \mathbf{v}_n.$$

The vector

$$[\mathbf{v}]_E = (c_1, c_2, \dots, c_n)^T$$

is called the coordinate vector of  $\mathbf{v}$  with respect to  $E$ .

#### Remark 4.1: Interpretation

The coordinate vector  $[\mathbf{v}]_E$  represents  $\mathbf{v}$  in  $\mathbb{R}^n$ , even if  $\mathbf{v}$  itself belongs to an abstract vector space  $V$ . This allows abstract vector problems to be converted into matrix problems.

## 5 Summary

### Summary

- A change of basis converts vectors from one coordinate system to another.
- Coordinates depend on the ordering of the basis.
- Transition matrices map coordinates between bases.
- If  $U$  is a transition matrix, then  $U^{-1}$  reverses the change of basis.
- Every finite-dimensional vector space admits coordinate representations via ordered bases.